Inter Quartile Range Calculation

Interquartile Range(IQR)

a. The interquartile range. Compare the two interquartile ranges.

b. Any outliers in either set.

The five number summary for the day and night classes is

	Minimum	Q_1	Median	Q_3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

ANSWER

INTER QUARTILE RANGE CALCULATION:

	Min	Q_1	Q_2	Q_3	Max
1) DAY	32	56	74.5	82.5	99.

$IQR = Q_3 - Q_1$

$= 82.5 - 56$

$IQR = 26.5$

Lesser range $= Q_1 - 1.5 \times IQR$

$= 56 - (1.5 \times 26.5)$

$= 56 - 39.75$

$= 16.25$

∴ Minimum Value Given is 32

So, no lesser out lies.

* Higher range $= Q_3 + 1.5 \times IQR$
$= 82.5 + (1.5 \times 16.25)$
$= 82.5 + 39.75$
$= 122.25$

Max value given is 99
So, no outlier in Day in higher range.

2) Night

$IQR = Q_3 - Q_1$
$= 89 - 78$
$= 11$

* Lesser range $= Q_1 - (1.5 \times IQR)$
$= 78 - (1.5 \times 11)$
$= 78 - 16.5$
$= 61.5$

Lesser value or Minimum value in Night is 25.5

$\therefore$ There is outlier in lesser range

* Higher range $= Q_3 + (1.5 \times IQR)$
$= 89 + (1.5 \times 11)$
$= 89 + 16.5$
$= 105.5$

$\therefore$ Max value $= 98$, No outliers in higher range.